

Appendix 1

Details Answers for Schedule A - Technical Req 6.0

Compute:

Google Cloud Platform (GCP) Compute Engine offers a wide range of virtual machine (VM) instances to meet diverse workload needs. This response outlines the configurable aspects related to processors, memory, and operating systems. Please note that specific availability of processor types, speeds, and other features may vary across regions and zones. Google continuously updates and expands its offering to take advantage of new technologies.

1. Cloud Regions:

- GCP operates across numerous regions globally, each containing multiple zones. For a complete and updated list of locations, please refer to the official Google Cloud documentation: <https://cloud.google.com/about/locations>.
- Region selection is critical: It impacts latency, cost, and compliance requirements. The specific region for our deployment would be determined based on the project's geographical requirements.
- Example Regions:
 - us-central1 (Iowa, USA)
 - europe-west1 (Belgium)
 - asia-east1 (Taiwan)
 - australia-southeast1 (Sydney, Australia)

2. Processor Family and Type:

- GCP offers instances based on processors from several vendors, categorized into families with varying performance characteristics.
- Processor Families:
 - Intel Xeon: Offers a balance of performance, cost, and broad compatibility. Often found in General-Purpose and Compute-Optimized instances.
 - AMD EPYC: Provides strong performance with a focus on core counts, often used in Memory-Optimized and Compute-Optimized instances.
 - Arm-based (Ampere Altra): Designed for performance and power efficiency. Well-suited for specific workloads.
 - Google's own processors (TPUs): Designed specifically for Machine Learning tasks and optimized for tensor processing. These are not general-purpose processors.
- Processor Types (Examples within each family):
 - Intel: Cascade Lake, Ice Lake, Sapphire Rapids
 - AMD: Milan, Genoa
 - Arm: Altra

3. Processor Speed:

- GCP provides the relative performance of various instance types through machine type categories. Clock speeds can vary based on the technology used by the processors. However, instances are optimized based on their designed purpose.
- It's more appropriate to consider the overall performance characteristics of an instance rather than precise clock speeds.
- Performance is influenced by:
 - Processor architecture (e.g., number of cores, cache sizes).
 - Turbo Boost technology

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- Specific workloads (e.g. single threaded vs multithreaded)

4. Processor and Core Counts:

- Processor Count: GCP virtual machines only have access to a single processor. This is the underlying physical processor the virtual machine uses.
- Core Counts: GCP offers instances with varying numbers of virtual CPUs (vCPUs), which are implemented as hyperthreads on the underlying physical cores.
 - Core count range: Ranges from as little as 1 core up to 224 cores per instance depending on the instance family.
 - Customization: Some instance families allow custom core counts within defined ranges.
- Hyperthreading Most cores support hyperthreading. The precise implementation and capabilities vary depending on the instance family.
 - vCPU's A vCPU is one thread on a processor core.
 - Note: In the context of the cloud provider's terminology, it is common to refer to a virtual core as a "vCPU".

5. Total or Per Core Memory Quantity (GB):

- Memory: GCP provides a range of memory options, directly related to the instance type:
 - General-Purpose: These instance types have memory ratios ranging from 3.75GB up to 8GB per vCPU
 - Memory-Optimized: These instances have high memory ratios, ranging from 12GB to 88GB per vCPU.
 - Compute-Optimized: These instances have lower memory ratios, 1GB to 2GB per vCPU.
- Customizable: Some instance families allow for customized memory allocations within defined ranges.

6. Supported and Excluded Operating System Families and Versions:

- Supported Operating System Families: GCP supports a wide range of operating systems, including:
 - Linux Distributions:
 - Debian
 - Ubuntu
 - CentOS/Red Hat Enterprise Linux (RHEL)
 - SUSE Linux Enterprise Server (SLES)
 - Container-Optimized OS (COS)
 - Rocky Linux
 - Alma Linux
 - Windows Server: Multiple versions are supported.
- Supported Versions: Specific versions are regularly updated. For the latest list, please consult: <https://cloud.google.com/compute/docs/images>.
- Excluded OS:
 - GCP does not provide support for legacy operating systems that have reached end-of-life. It is important to use operating system versions that are actively supported by their vendors to be secure and benefit from ongoing maintenance.
 - Operating systems not in the officially supported list can be installed with the understanding that full compatibility and technical support might not be available.
- Operating System Images:
 - GCP provides public images which are configured and maintained.

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- You can create your own custom images for more specialized workloads.
- Marketplace Images provide operating systems and application packages from 3rd party vendors.

Storage:

GCS is a highly scalable, durable, and secure object storage service designed for a wide range of data storage needs, from backups and archives to web content and analytics.

1. Cloud Region:

- Global and Regional Options: GCS offers both regional and multi-regional storage classes, providing flexibility to balance performance and cost.
 - Regional: Data is stored within a specific geographic region (e.g., us-central1, europe-west1). Best for low-latency access and lower storage costs.
 - Multi-Regional: Data is replicated across multiple geographic regions within a larger area (e.g., us, eu). Best for high availability and global access.
 - Dual-Region: Data is replicated across two specific regions. Offers high availability with some locality for lower latency.
- Availability: GCS is available in numerous regions worldwide, allowing you to choose the best location(s) based on your needs.
- Data Residency: GCP allows for very specific control over the region in which your data resides, assisting with compliance requirements.

2. Storage Type:

- Object Storage: GCS is fundamentally an object storage service. This means data is stored as individual objects (files) along with associated metadata.
- Storage Classes: GCS provides different storage classes optimized for different use cases:
 - Standard: For frequently accessed ("hot") data.
 - Nearline: For data accessed less frequently (e.g., monthly).
 - Coldline: For infrequently accessed data (e.g., quarterly or annually).
 - Archive: For long-term archival data with very low access requirements.

3. Connection Type:

- HTTPS: The primary connection method for accessing GCS is via secure HTTPS REST APIs. This provides encrypted data transfers.
- Client Libraries: Google provides client libraries for various programming languages (e.g., Python, Java, Go, .NET) that make interacting with GCS straightforward.
- Command-Line Interface (CLI): GCP offers the gsutil CLI tool for managing GCS buckets and objects.
- Cloud Console: You can interact with GCS through the GCP Cloud Console's web interface.
- Integrations: GCS integrates with other GCP services (e.g., Compute Engine, Dataproc, Dataflow) and also third party applications and services via APIs.

4. Read and Write Speeds:

- Scalable Performance: GCS is designed for high performance and can handle large amounts of data and concurrent access.
- Performance Factors: Read and write speeds are influenced by:

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- Storage Class: Standard class has the highest performance; lower classes have latency trade-offs.
 - Network Bandwidth: The network connection to GCS and the client's network speeds affect performance.
 - Object Size: Larger objects are often written with higher throughput.
 - Multi-threading: Parallel reads/writes are very effective for enhancing throughput.
 - Location: Location of client relative to GCS bucket.
 - No Guaranteed Speeds: Specific read/write speeds are not guaranteed because of these factors. Instead, GCS is built to provide optimal performance based on available resources.
 - High Performance: Throughput can easily reach multi gigabits per second.
5. High Availability/Redundancy Service Level:
- High Durability: GCS is designed for extreme durability: 99.999999999% (eleven 9's) annual durability across all storage classes.
 - Redundancy: Data is replicated across multiple availability zones (and in the case of multi-regions, across regions), ensuring data remains accessible even if an availability zone or data center has a failure.
 - Availability Guarantees: Google provides Service Level Agreements (SLAs) with specific availability guarantees for different storage classes:
 - Standard Class: 99.95% Availability within a single-region and 99.99% Multi-region.
 - Nearline, Coldline, and Archive: 99.9% Availability within a single-region and 99.99% Multi-region.
 - Geographic Redundancy: Multi-regional and dual-regional classes provide geographic redundancy which provide the highest levels of availability.
6. Data Backup and Recovery Service Level:
- Automatic Replication: GCS automatically replicates your data for durability. This can effectively serve as backup.
 - Versioned Buckets: Enable versioning on buckets to keep previous versions of objects in case of accidental deletion or modification.
 - Lifecycle Management: GCS offers lifecycle policies that can automatically transition objects between storage classes or delete them based on time or rules.
 - Object Replication Policy: You can also set up an Object Replication Policy to copy objects to another region within the same project or even a separate project. This acts as both a backup and a disaster recovery mechanism.
 - Recovery Process: Data recovery is primarily done by retrieving the required objects directly from GCS. There is no separate service or action that needs to be performed to perform a recovery as long as the required data has not been deleted.
 - Snapshot: Snapshots are not directly available on GCS like some other block storage services. Object versioning and Replication policies offer analogous functions.
7. Ingress, Egress, and Bulk Data Migration Support:
- Ingress:
 - API Access: Data is uploaded to GCS primarily through its REST API.
 - gsutil: The command line tool is well equipped for file uploads.
 - Web Console: Files can be uploaded via the console.
 - Data Transfer Service: For transferring large data sets from external sources (e.g., AWS S3, on-prem), GCP offers the Data Transfer Service (DTS).

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- Transfer Appliance: For very large offline bulk transfers, Google can ship a hardware appliance to load data to and ship it back for upload.
- Egress:
 - API Access: Data is downloaded from GCS using the REST API and other access methods.
 - gsutil: The command line tool is well equipped for file downloads.
 - Web Console: Files can be downloaded via the console.
 - Data Transfer Service: DTS can also be used to export data from GCS to other providers or on-premises.
- Bulk Data Migration:
 - Parallel Transfers: GCS supports and encourages parallel transfers for maximum performance.
 - Data Transfer Service: DTS excels at migrating bulk data, using managed and reliable transfer operations.
 - Transfer Appliance: Physical appliance for bulk data uploads.
 - gsutil: Suitable for many bulk transfers when done in parallel.
- Data Movement: Data movement, for example, between buckets in GCS can be done using the gsutil tool or the API. You do not need to move the data through a local host machine.

Network:

Google Cloud Platform (GCP) provides a robust and scalable network infrastructure designed to meet the demanding needs of modern applications.

1. Sustained and Burst Circuit Speeds (Upload & Download):

- GCP's network is designed for high performance and uses a combination of technologies to achieve both sustained and burst capacity:
- Egress (Download) Speeds:
 - Sustained: GCP's egress (data leaving GCP) bandwidth is generally governed by the tier and size of your Compute Engine instances or other cloud services. Larger instances generally have higher egress capacity. Sustained throughput is typically well-optimized and predictable.
 - Burst: GCP can handle significant bursts of traffic. However, the burst limit is dependent on the overall network capacity, region, zone, and specific instance type. While it's not explicitly a guaranteed "burst" limit in terms of a specific Mbps/Gbps, the infrastructure can handle bursts that exceed sustained throughput for brief periods.
 - Example: A n2-standard-16 compute instance might have a sustained egress bandwidth in the ~10Gbps range but can likely handle higher throughput for shorter bursts.
- Ingress (Upload) Speeds:
 - Sustained: Ingress (data entering GCP) is generally considered highly reliable and fast within the Google Network. Sustained speeds depend upon the source network, the connection being used, and also the network interfaces of your Virtual machines.
 - Burst: Similar to egress, ingress can also handle traffic bursts. GCP's global network is engineered to handle high volumes of ingress data efficiently.

2. Physical and Virtual Dedicated Connections:

GCP offers two primary dedicated connection options:

- Cloud Interconnect (Physical Dedicated):

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- Description: Cloud Interconnect provides dedicated, high-bandwidth connections between your on-premises network and your Virtual Private Cloud (VPC) in Google Cloud. This bypasses the public internet, offering lower latency, more consistent performance, and improved security.
- Options:
 - Dedicated Interconnect: Connect directly to Google's edge locations. Offers very high bandwidth (10Gbps or 100Gbps) and the lowest latency.
 - Partner Interconnect: Connect through a supported third-party partner who is already connected to Google's network. Offers flexible bandwidth options (50Mbps to 10Gbps) and is suitable for smaller deployments.
- Features:
 - Consistent Performance: Predictable latency and high bandwidth compared to internet connections.
 - Enhanced Security: Bypasses the public internet, minimizing exposure to security threats.
 - Direct Connections: Creates a direct link between your network and Google's network.
- Virtual Dedicated Connections (VPN):
 - Description: While not a "physical" connection, Cloud VPN creates a secure and encrypted tunnel over the internet to connect your on-premises network to GCP.
 - Options:
 - Classic VPN: An IPsec VPN connection. Provides secure encrypted connectivity over the public internet.
 - HA VPN: A higher availability VPN. Uses multiple gateways and VPN tunnels across multiple regions for maximum resilience.
 - Features:
 - Secure Communication: Provides secure, encrypted communication over the internet.
 - Flexibility: Easier to deploy and scale than physical connections.
 - Cost-Effective: Often more cost-effective than dedicated physical connections.

3. Included and Excluded Network Features:

Included Network Features (Within VPC):

- Virtual Private Cloud (VPC): GCP's fundamental network layer allows you to create logically isolated networks within Google Cloud.
- Subnets: Divide your VPC into smaller, manageable subnets.
- Routes: Define how traffic is routed within your VPC and to external networks.
- Firewall Rules: Control network access by allowing or blocking specific traffic based on source and destination.
- Internal Load Balancers: Distribute traffic across instances within your VPC for higher availability and scalability.
- Cloud DNS: Host your domain names and manage your DNS records.
- Cloud NAT: Allow your private instances to access the internet without public IP addresses.

Excluded Network Features (Generally Require Add-On Services or Configurations):

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- **Advanced Threat Detection and Prevention:** For more sophisticated security inspection, you may need additional services such as Google Cloud Security Command Center or third-party network security tools.
- **Deep Packet Inspection (DPI):** GCP doesn't directly offer DPI, but it can be achieved with partner solutions or custom network appliances.
- **Complex Routing Policies:** For more complex routing scenarios, you might need to use Cloud Router with Border Gateway Protocol (BGP).
- **Network Segmentation & Microsegmentation:** Beyond basic subnetting and firewalling, advanced segmentation requires planning, custom configurations, and often third-party tools.
- **Shared VPC:** If you want to share a VPC across multiple projects, you'll need to set up Shared VPC.

4. Associated Costs:

GCP Network Costs are generally based on:

- **Data Egress:** Data going out of GCP to the Internet, to your on-premises locations or to other cloud providers.
- **Inter-region and inter-zone Traffic:** Data transferred between different regions or availability zones.
- **Cloud Interconnect:** Hourly port fees for dedicated connections and any data transfer through those connections.
- **Cloud VPN:** Hourly VPN gateway fees and data transfer through VPN tunnels.
- **Public IP Addresses:** Charges for reserved static IP addresses.

GCP Relational Databases

1. Cloud Region:

- **Global Availability:** GCP has a wide network of regions and zones across the globe.
- **Region Selection:** Clients should select a region that is geographically closest to their user base to minimize latency and improve performance. Examples include:
 - us-central1 (Iowa, USA)
 - us-east1 (South Carolina, USA)
 - europe-west1 (Belgium)
 - asia-east1 (Taiwan)
 - asia-southeast1 (Singapore)
 - australia-southeast1 (Sydney)

A full list of available regions is available on the GCP website

2. Database Family and Type:

GCP offers several relational database options, categorized as follows:

- **Cloud SQL:** Fully managed relational database service for:
 - **MySQL:** Widely used open-source database.
 - **PostgreSQL:** Powerful and versatile open-source database.
 - **SQL Server:** Microsoft's enterprise database solution.
- **Spanner:** Globally distributed, scalable, and strongly consistent database.
- **AlloyDB for PostgreSQL:** PostgreSQL-compatible database with advanced features.

3. Processor or Core Count:

- **Cloud SQL:**

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- Configurable vCPU counts, varying depending on the database type and tier selected. Common configurations are:
 - Shared-core: For low-cost development environments.
 - Dedicated core: For production environments.
 - Spanner:
 - Units are called processing units (PUs) for compute, and the exact core count depends on the underlying infrastructure used by Spanner.
 - Scales horizontally based on data volume and processing needs.
 - AlloyDB for PostgreSQL:
 - Configurable vCPU counts. Similar to Cloud SQL dedicated-core instances.
4. Memory Quantity in GB:
- Cloud SQL:
 - Memory (RAM) is configurable and directly tied to the selected vCPU count and tier. Common memory configurations are:
 - Spanner:
 - Memory is provisioned automatically, based on the needs of the workload and scaling capacity of PUs.
 - AlloyDB for PostgreSQL:
 - Memory (RAM) is configurable and directly tied to the selected vCPU count.
5. High Availability/Redundancy Service Level:
- Cloud SQL:
 - Regional: Instances can be deployed in a highly available (HA) setup with automatic failover between zones within the same region. HA typically includes:
 - Automatic failover to a standby instance in another zone within the same region.
 - Replicated storage
 - Maintenance and updates.
 - Spanner:
 - Multi-Regional and Global: Inherently highly available and redundant across multiple regions or globally, based on the configuration.
 - Offers 99.999% availability for multi-regional deployments.
 - AlloyDB for PostgreSQL:
 - Regional HA: Instances can be deployed in a highly available (HA) setup with automatic failover between zones within the same region.
6. Data Backup and Recovery Service Level:
- Cloud SQL:
 - Automated Backups: Configurable automated daily backups are available with configurable retention periods.
 - On-Demand Backups: Manual backups can be created on demand.
 - Point-in-Time Recovery (PITR): Recovery to a specific point in time is available for accidental data loss or corruption.
 - Spanner:
 - Automatic Backups: Continuous backups are automatically performed for data protection.
 - Point-in-Time Recovery (PITR): Recovery to a specific point in time within the retention window is available.

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- Schema Changes: Schema changes can be rolled back.
- AlloyDB for PostgreSQL:
- Automated Backups: Configurable automated daily backups are available with configurable retention periods.
- On-Demand Backups: Manual backups can be created on demand.
- Point-in-Time Recovery (PITR): Recovery to a specific point in time is available for accidental data loss or corruption.

Containers:**1. Cloud Regions:**

- GCP offers a global network of cloud regions, each with multiple availability zones. You can deploy containers in the following regions.
- A detailed list of available regions and zones can be found at:
<https://cloud.google.com/compute/docs/regions-zones>
- Some popular regions include:
 - us-central1 (Iowa, USA)
 - us-east1 (South Carolina, USA)
 - us-west1 (Oregon, USA)
 - europe-west1 (Belgium)
 - asia-east1 (Taiwan)
 - asia-northeast1 (Tokyo)
 - australia-southeast1 (Sydney)

2. Clusters (Kubernetes Engine):

- Google Kubernetes Engine (GKE): Our primary container orchestration platform is GKE, a managed Kubernetes service.
- Cluster Types:
 - Standard: Offers full flexibility and control over the Kubernetes master and node configuration.
 - Autopilot: Provides a fully managed cluster experience with simplified node management, scaling, and maintenance, optimized for cost and operational efficiency.
- Node Pools: Clusters are comprised of node pools, which are groups of identical virtual machines (instances) that host the containers.
- Node Scaling: GKE supports autoscaling based on workload demand.

3. Instance Series (Compute Engine):

- GKE nodes leverage Compute Engine instance families.
- General-Purpose:
 - e2 Series: Cost-optimized for a variety of workloads.
 - n2 Series: Balanced compute, memory, and networking.
 - n2d Series: Based on AMD processors, cost-effective and high-performance.
 - n1 Series: Older generation, still available for compatibility, but newer generations are preferred.
- Compute-Optimized:
 - c2 Series: High-performance for compute-intensive tasks.
 - c2d Series: AMD-based high performance.

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- Memory-Optimized:
 - m1 Series: Designed for large memory workloads.
 - m2 Series: Next-generation, larger memory options.
 - GPU:
 - a2 Series: Powered by NVIDIA A100 GPUs.
 - g2 Series: Powered by NVIDIA L4 GPUs.
 - t4 Series: Powered by NVIDIA T4 GPUs (cost effective)
 - Note: The specific availability of instance series can vary by region and zone.
 - A detailed list of available instance series can be found at: <https://cloud.google.com/compute/docs/machine-types>
4. Supported & Excluded Operating System Families and Versions (GKE Nodes):
- Supported Operating Systems:
 - Container-Optimized OS (COS): Google's actively maintained and highly optimized OS designed for running containers, is the preferred option.
 - Ubuntu: Widely used Linux distribution.
 - Windows Server (For Windows node pools): Support for running Windows-based containers.
 - Other Linux distributions: May be available as community images, but officially supported options are recommended.
 - Supported Versions: GKE typically supports the latest stable and several previous versions of these OSes, with ongoing maintenance and security updates provided. Check the latest GKE release notes for detailed version support: <https://cloud.google.com/kubernetes-engine/docs/release-notes>
 - Excluded Operating Systems: Generally, older, end-of-life versions of operating systems are not supported for security reasons. Also, specific less common OSes may be excluded if not optimized for container workloads.
5. Managed OS Disks in GB:
- Disk Type: GKE nodes default to using Persistent Disks (block storage).
 - Managed OS Disk Sizes: The size of the OS disk for a GKE node is managed and automatically configured by GKE based on the instance type.
 - Minimum Disk Size: Generally, node pools have a default disk size of at least 100GB. The actual size might vary slightly based on the instance family and OS selection.
 - Customization: While you cannot directly configure the default OS disk size, you can adjust the size and type of disks (local SSDs, standard HDD) mounted for application data using persistent volumes and storage classes.
 - Local SSDs Can be added for faster performance but are ephemeral.

Data Warehouse Services:**1. BigQuery**

- Cloud Region:
 - Global (data resides in chosen regions) or specific regions depending on the dataset and its data location requirements. GCP offers a wide variety of regions around the globe. You must specify a single location to store data.
- Database Family and Type:
 - Family: Data Warehouse

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- Type: Serverless, fully managed analytical data warehouse.
- Processor or Core Count:
 - Serverless; no user specification of CPU cores. Automatically scales based on query complexity and workload. Resources are abstracted from the user. The user is billed based on usage.
- Memory Quantity (GB):
 - Serverless; no user specification of memory. Automatically scales based on query requirements. User is billed based on usage.
- High Availability/Redundancy Service Level:
 - High Availability: Built-in with no user configuration or management. Service is available if one availability zone fails, it'll continue to operate from other zone in the same region.
 - Redundancy: Data is automatically replicated across multiple availability zones within the chosen region to ensure resilience.
- Data Backup and Recovery Service Level:
 - Backup: Automatic, continuous backup of metadata.
 - Recovery: Point-in-time data recovery within a 7-day window (standard). Time-travel enables users to retrieve data and reconfigure based on previous versions. Longer retention periods are available.
 - Users can also export and backup to Cloud Storage.

2. Cloud Spanner

- Cloud Region:
 - Supports multi-region (strongly consistent) or regional configurations. Spanner has locations around the globe that are constantly expanding. Multi-region configurations enable multiple locations across the globe.
- Database Family and Type:
 - Family: Relational Database
 - Type: Globally-distributed, scalable, strongly consistent relational database.
- Processor or Core Count:
 - Configurable in terms of "processing units" which correlate to CPU cores, memory and network bandwidth. Users define the number of nodes when creating instance of spanner. Each node allocates processing units. Processing units per node vary based on configuration.
- Memory Quantity (GB):
 - Scalable based on the number of nodes and processing units chosen. Memory is not directly allocated or configured. Storage allocation is separate.
- High Availability/Redundancy Service Level:
 - High Availability: Built-in with multi-regional instances for cross region and availability zone failover, and automatic failovers. Service is highly resilient.
 - Redundancy: Data is automatically replicated across multiple availability zones within a single region and across regions if deployed with multi-region configuration. Data is stored in multiple fault-tolerant replicas.
- Data Backup and Recovery Service Level:
 - Backup: Supports automated or manual backups.
 - Recovery: Allows point-in-time recovery to restore from backups or previous points in time. Restore is from both Cloud Storage and Cloud Spanner.

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- Backup retention can be configured.
3. Cloud SQL (PostgreSQL, MySQL, SQL Server)
- Cloud Region:
 - Regional instances for high performance, or multi-region configurations for cross regional failover.
 - Database Family and Type:
 - Family: Relational Database
 - Type: Managed relational database services. Supports various engine types: PostgreSQL, MySQL, SQL Server.
 - Processor or Core Count:
 - Configurable based on the selected machine type (e.g., standard or high-memory).
 - Memory Quantity (GB):
 - Configurable based on the selected machine type.
 - High Availability/Redundancy Service Level:
 - High Availability: Provides options for high availability with automatic failover. Can be deployed with either zonal or regional disaster recovery and failover.
 - Redundancy: Data is replicated across multiple zones in the region.
 - Data Backup and Recovery Service Level:
 - Backup: Automated backups, configurable backup scheduling.
 - Recovery: Point-in-time recovery from backups. Recovery from backups or through clones of a previous instance.
4. Dataproc (Hadoop/Spark)
- Cloud Region:
 - Regional clusters. Data must be stored in the same region as the Dataproc cluster.
 - Database Family and Type:
 - Family: Data Processing
 - Type: Managed Hadoop and Spark service. Not a database, but often used in conjunction with data warehouses.
 - Processor or Core Count:
 - Configurable through the selection of machine types and the number of worker nodes.
 - Memory Quantity (GB):
 - Configurable through the selection of machine types and number of worker nodes. Memory for the workers nodes is configurable and allocated during creation.
 - High Availability/Redundancy Service Level:
 - High Availability: Can be configured for HA master nodes in different zones.
 - Redundancy: Data is usually stored in Cloud Storage, which is highly available and redundant. Workers are deployed across zones.
 - Data Backup and Recovery Service Level:
 - Backup: Backup and recovery is not part of Dataproc. Data is stored separately on Cloud Storage and is backed up according to cloud storage policies.
 - Recovery: Recovery involves re-creating the cluster and running jobs against the data in Cloud Storage.

Functions as a service:

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Google Cloud Functions provides a powerful, serverless platform for building and deploying event-driven applications. Its flexible architecture, automatic scaling, and high availability make it suitable for various use cases. While specific features like WAF and DPI require integration with other GCP services, GCF greatly simplifies the operational burden of running web services and other applications.

1. Cloud Region:

- **Flexibility:** GCF is available in numerous Google Cloud regions globally. This allows you to deploy your functions closer to your users, minimizing latency.
- **Selection:** You can choose the specific region when deploying your function. Consider factors like:
 - **User Location:** Deploy in the region closest to the majority of your users for the best performance.
 - **Data Sovereignty:** Select regions that meet your data residency and compliance requirements.
 - **Resource Availability:** Some regions might have specific hardware or service variations.

2. Memory Quantity:

- **Configurable:** You can allocate memory to your GCF function, which dictates resources and influences performance.
- **Range:** The memory options range from 128 MB to 8 GB, adjustable in 256 MB increments.
- **Impact:**
 - **Performance:** More memory can improve execution speed and allow your functions to process larger datasets.

3. Connection Speed:

- **High-Speed Network:** GCF leverages Google's high-speed global network for both inbound requests and outbound connections.
- **Low Latency:** The underlying infrastructure is designed for low latency and high throughput, ensuring fast execution.
- **Dynamic Scaling:** The network bandwidth dynamically scales to meet the needs of your function, handling traffic spikes effectively.
- **Inter-Service Communication:** GCF uses Google's internal, high-speed network for communication with other Google services and within a VPC.

4. High Availability/Redundancy Service Level:

- **Managed Infrastructure:** GCF is a fully managed service, meaning Google handles the underlying infrastructure for high availability and redundancy.
- **Multi-AZ Deployment:** Your functions are automatically deployed across multiple availability zones (AZs) within the selected region, providing redundancy.
- **Fault Tolerance:** If one AZ becomes unavailable, your functions will continue to run in the other available AZs without any interruption.
- **Automatic Scaling:** GCF automatically scales your functions based on the number of incoming requests, ensuring high availability even during traffic spikes.
- **SLA:** Google offers a Service Level Agreement (SLA) that guarantees a certain level of service availability (typically 99.9%).
- **Consideration:** You can expect a highly reliable service with minimal downtime due to the built-in redundancy and auto-scaling features.

5. Included/Excluded Web Service Features:

Included:

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- Automatic Scaling: GCF scales automatically based on traffic demand.
- Serverless Execution: No server management required, eliminating the operational overhead.
- Trigger-Based Execution: Functions are triggered by events like HTTP requests, Cloud Storage changes, Pub/Sub messages, Cloud Firestore writes, etc.
- Language Support: Supports a variety of languages such as Python, Node.js, Go, Java, .NET, Ruby and PHP.
- Libraries: Supports a wide variety of libraries through package management (npm, pip, maven, etc.), including most common HTTP, Database and other third-party libraries
- Environment Variables: Configurable environment variables for application configuration.
- Logging and Monitoring: Integration with Cloud Logging and Cloud Monitoring for detailed insights.
- Cloud IAM Integration: Robust Identity and Access Management for secure access control.
- Private Network (VPC) Support: Option to deploy functions within your Virtual Private Cloud (VPC).
- HTTP/HTTPS Support: Function endpoints respond to HTTP/HTTPS requests
- Secrets Management: Integration with Secret Manager for managing sensitive data.
- Cloud Trace and Profiling: Tools for performance analysis and optimization.
- Authentication: Support for API Keys, Google Authentication, or other custom authentication solutions
- Version Management: Option to control versions of deployments for rolling updates and A/B testing
- Background Functions: Support for running asynchronous tasks in the background with no timeout restrictions
- Concurrency Control: Configure concurrency levels to optimize functions for different workload patterns

Excluded (Typically Managed by other GCP Services):

- Web Application Firewall (WAF): GCF itself doesn't include a WAF. You can use Cloud Armor in front of your function if it's an HTTP function.
- SSL/TLS Certificate Management: GCF automatically uses HTTPS; you do not handle SSL certificate management.
- Custom DNS Configuration: You cannot use GCF with custom DNS names directly. You will need to use a load balancer like Cloud Load Balancer and other services to point your custom domain name to your GCF function URL
- Traffic Inspection (DPI): GCF does not provide deep packet inspection capabilities. You will have to integrate other services for these specific use cases.
- Content Delivery Network (CDN): You can connect a GCF function to Cloud CDN for improved delivery of static assets if the GCF function handles the retrieval.
- Database Server Management: While GCF can connect to databases like Cloud SQL, it doesn't manage the database servers themselves.
- Dedicated IP Addresses: GCF functions typically run on shared infrastructure and do not come with dedicated static IP addresses. Use Cloud NAT for static outbound IP addresses.

6. Associated Costs:

- Pay-per-Use: GCF is a pay-per-use service. You are charged only for the time your function is actively executing.
- Pricing Metrics: Costs are based on:

Appendix 1

Details Answers for Schedule A - Technical Req 6.0

- Execution Time (CPU Time): Time spent in running your function.
 - Memory Allocation: The amount of memory allocated to your function.
 - Number of Invocations: The number of times your function is called.
 - Networking: Egress traffic costs for outgoing requests (minimal if within GCP).
- Free Tier: Google offers a free tier for GCF, which may be sufficient for experimentation.
- Cost Optimization: We can help you optimize your function code to improve execution time and reduce cost.
- Budget Management: You can set budget limits and configure alerts to control costs.